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(2013.01)(21) Appl. No.: **18/861,747**(57) **ABSTRACT**(22) PCT Filed: **May 24, 2023**(86) PCT No.: **PCT/EP2023/063857**

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The present invention relates to a process for producing middle distillates from a feedstock comprising a residual hydrocarbonaceous feedstock and a hydrogen deficient feedstock. The residual hydrocarbonaceous feedstock is deasphalted to obtain a deasphalted product of which at least 50 wt % has a boiling point above 550° C. and an asphaltic product. The deasphalted product is combined with the hydrogen deficient feedstock to produce a mixed deasphalted product, which is hydrodemetallized. The hydrodemetallized product is hydrotreated and hydrocracked. The hydrocracked product is subjected to a separation treatment to produce at least a middle distillate fraction.

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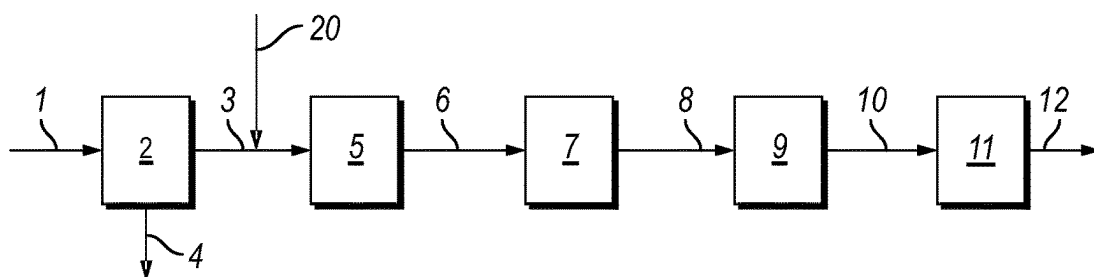


FIG. 1

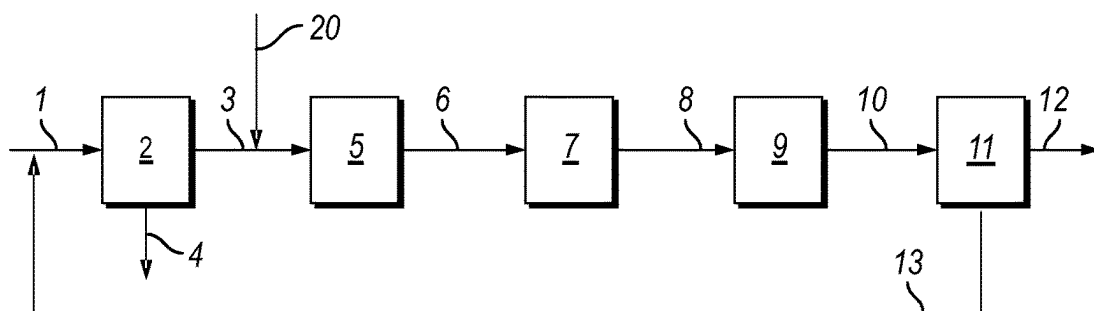


FIG. 2

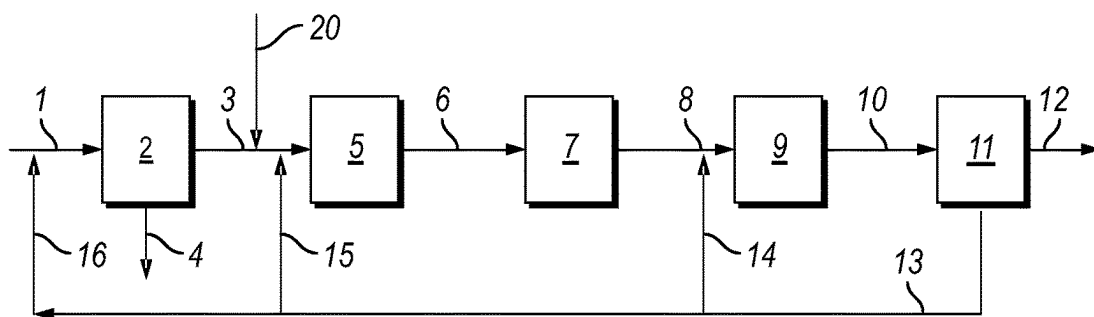


FIG. 3

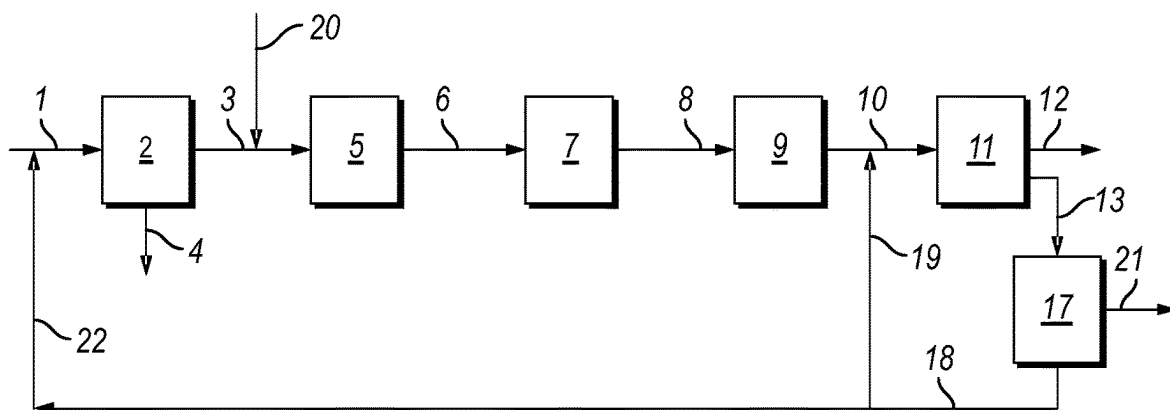


FIG. 4

PROCESS FOR PRODUCING MIDDLE DISTILLATES

FIELD OF THE INVENTION

[0001] The present invention relates to process for producing middle distillates from residual hydrocarbonaceous feeds and hydrogen deficient feeds.

BACKGROUND OF THE INVENTION

[0002] In view of increasing environmental awareness which desires to reduce the carbon footprint of processes, much research and development is directed to increasing yields of middle distillates and also processing of low hydrogen content feeds which would increase the production of ultra low sulphur middle distillates such as ultra low sulphur diesel fuels which may include some portion of renewable origin.

[0003] It is known to produce ultra low sulphur diesel fuels by first hydrodesulphurising a hydrocarbon distillate stream boiling in the gasoil boiling range and then catalytically dewaxing the desulphurised distillate stream. Especially in the winter time the catalytic dewaxing step may be needed for removing waxy molecules from the distillate stream in order to reduce the cloud point and the pour point of the gasoil. The desulphurised and dewaxed gasoil may be hydrofinished for saturation of aromatic compounds. In this way the cetane index or number of the gasoil product may be further enhanced. The resulting desulphurised, dewaxed and optionally hydrofinished gasoil is then used as diesel fuel or diesel fuel component.

[0004] Objects of the present invention are to provide a process for producing ultra low sulphur middle distillates in high yields and incorporating hydrogen deficient feeds.

SUMMARY OF THE INVENTION

[0005] This object is achieved when use is made of a particular multi-step process. Accordingly, the present invention relates to a process for producing middle distillates from a feedstock comprising a residual hydrocarbonaceous feedstock and a hydrogen deficient feedstock, comprising the steps of:

[0006] (a) deasphalting the residual hydrocarbonaceous feedstock to obtain a deasphalted product of which at least 50 wt % has a boiling point above 550° C. and an asphaltic product;

[0007] (b) combining the deasphalted product with the hydrogen deficient feedstock to produce a mixed deasphalted product, wherein the hydrogen deficient feedstock has a hydrogen (H) content of at least 6 wt % to at most 11.3 wt %;

[0008] (c) hydrodemetallizing at least part of the mixed deasphalted product from step (b) to produce a hydrodemetallized product;

[0009] (d) hydrotreating at least part of the hydrodemetallized product from step (c) to produce a hydrotreated product;

[0010] (e) hydrocracking at least part of the hydrotreated product from step (d) to produce a hydrocracked product; and

[0011] (f) subjecting at least part of the hydrocracked product from step (e) to a separation treatment to produce at least a middle distillate fraction.

[0012] In accordance with the present invention high yields of middle distillates containing less than 10 ppm wt sulphur can advantageously be produced from residual hydrocarbonaceous feedstocks and hydrogen deficient feedstock.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is a simplified block flow diagram of an embodiment of a process for producing middle distillates from residual hydrocarbonaceous feeds and hydrogen deficient feeds.

[0014] FIG. 2 is a simplified block flow diagram of alternate embodiment of a process for producing middle distillates from residual hydrocarbonaceous feeds and hydrogen deficient feeds.

[0015] FIG. 3 is a simplified block flow diagram of alternate embodiment of a process for producing middle distillates from residual hydrocarbonaceous feeds and hydrogen deficient feeds.

[0016] FIG. 4 is a simplified block flow diagram of alternate embodiment of a process for producing middle distillates from residual hydrocarbonaceous feeds and hydrogen deficient feeds.

DETAILED DESCRIPTION OF THE INVENTION

[0017] The residual hydrocarbonaceous feedstocks to be used in accordance with the present invention can be residual hydrocarbon oils, such as those obtained in the distillation of crude oils at atmospheric or reduced pressure. Suitably, at least 55 wt %, preferably at least 75 wt %, more preferably at least 85 wt %, and even more preferably at least 90 wt % of the residual hydrocarbonaceous feedstock has a boiling point of above 550° C. Atmospheric residues or vacuum residues contain however considerable amounts of non-distillable compounds having a high molecular weight such as asphaltenes. It is therefore considered desirable to remove asphaltenes from a residual hydrocarbon oil feed prior to subjecting the residual hydrocarbon oil to subsequent upgrading steps.

[0018] In step (a), a residual hydrocarbonaceous feedstock is deasphalted to obtain a deasphalted product of which at least 50 wt %, preferably at least 70%, more preferably at least 80 wt %, and even more preferably at least 85 wt % has a boiling point above 550° C. and an asphaltic product.

[0019] The deasphalting in step (a) may be carried out in any conventional manner. A well known and suitable deasphalting method is solvent deasphalting. In accordance with the present invention the deasphalting in step (a) is preferably carried out by means of a solvent deasphalting treatment.

[0020] In solvent deasphalting the hydrocarbon feed is treated counter-currently with an extracting medium which is usually a light hydrocarbon solvent containing paraffinic compounds. Commonly applied paraffinic compounds include C3-8 paraffinic hydrocarbons, such as propane, n-butane, isobutane, n-pentane, isopentane, hexane or mixtures of two or more of these. For the purpose of the present invention, it is preferred that C3-C5 paraffinic hydrocarbons, most preferably butane, pentane or a mixture thereof, are used as the extracting solvent. In general, the extraction depth increases at increasing number of carbon atoms of the extracting solvent. In this connection it is noted that the

higher the extraction depth, the larger the amount of hydrocarbons being extracted from the residual hydrocarbonaceous feedstock, the smaller and more viscous the asphaltic product will be, whereby the heavier the asphaltenes will be in the asphaltic product to be obtained in step (a).

[0021] In a solvent deasphalting treatment, a rotating disc contactor or a plate column can be used with the residual hydrocarbonaceous feedstock entering at the top and the extracting solvent entering at the bottom. The lighter hydrocarbons which are present in the residual hydrocarbonaceous feedstock dissolve in the extracting solvent and are withdrawn as the deasphalted product at the top of the apparatus. The asphaltenes which are insoluble in the extracting solvent are withdrawn in the form of the asphaltic product at the bottom of the apparatus. The conditions under which deasphalting takes place are known in the art. Suitably, deasphalting is carried out at a total extracting solvent to residual hydrocarbon oil ratio of 1.5 to 8 wt/wt, a pressure of from 1 to 60 bara and a temperature of from 40 to 200° C.

[0022] A deasphalting treatment generally causes a substantial amount of the metallic contaminants present in the feed as high-molecular weight complexes to accumulate in the asphaltic product rather than in the deasphalted product. Nonetheless, the metals content of the deasphalted product will be such that the deasphalted product needs to be subjected to a hydrodemetallizing step before it can be subjected to further hydroprocessing upgrading steps.

[0023] In step (b), at least part of the deasphalted product is combined with a hydrogen deficient feedstock to produce a mixed deasphalted product. Preferably, in step (b) the entire deasphalted product as obtained in step (a) is combined with the hydrogen deficient feedstock. The addition of the hydrogen deficient feedstock lowers the carbon footprint of the process by utilizing low value products from other processes and increasing the amount of middle distillate product.

[0024] In some embodiments, the mixed deasphalted product should have a hydrogen content less than at least 0.1 wt % of the deasphalted product via the addition of the hydrogen deficient feedstock. In other embodiments, the hydrogen content of the mixed deasphalted product should be at most 2 wt % lower than the deasphalted product via the addition of the hydrogen deficient feedstock. In some embodiments, the amount of hydrogen deficient feedstock comprises up to about 50 wt % of the mixed deasphalted product, or up to about 40 wt % of the mixed deasphalted product, or up to about 30 wt % of the mixed deasphalted product.

[0025] The deasphalted product from step (a) is a heavy deasphalted product. This means that at least 50 wt %, preferably at least 70 wt %, more preferably at least 80 wt %, and even more preferably at least 85 wt % of the deasphalted product to be treated in step (b) has a boiling point of above 550° C.

[0026] The hydrogen deficient feedstock has a hydrogen (H) content of at least 6 wt % to at most 11.3 wt % and a true-boiling point chromatography (TBP-GLC) distillation temperature of 10 wt % to exceed 280° C. Examples of such hydrogen deficient feedstock may be a slurry oil from a fluid catalytic cracker unit, tall oil pitch from the distillation of crude tall oil, ethylene cracker residue from either a fossil fuel and/or waste plastic feed. In some embodiments, the waste plastic feed may be treated by pyrolysis and/or hydroprocessing prior to being fed to the ethylene cracker.

[0027] Slurry oil is the bottom product from a Fluid Catalytic Cracker's (FCC) main distillation column and is a blending component for marine fuels. Slurry oil has a relative viscosity between 5 to 25 cSt@100° C and a total aromatics content between 35 and 65% wt. The hydrogen deficient slurry oil has a hydrogen content of about 8.2 to about 9.2% wt %. Slurry oil has TBP-GLC distillation temperature of 10 wt % of about 360° C. (680° F.).

[0028] Tall Oil Pitch (TOP) is a by-product of the distillation of Crude Tall Oil (CTO). One application of TOP today is as a bio heating fuel or as an alternative of fossil non-renewable energy. CTO is a by-product of the paper industry, which process wood pulp from pine trees. The hydrogen deficient TOP has a hydrogen content of about 10.7 to 11.1% wt TOP has a TBP-GLC distillation temperature of 10 wt % of at least 350° C. (662° F.). TOP has a viscosity at 70° C. ranging between 60 and 100 cSt. The use of TOP provides conversion of renewable pitch into valuable products, including automotive fuels. TOP is considered to be a renewable feed component and would provide carbon credit.

[0029] In step (c), at least part of the mixed deasphalted product as obtained in step (b) is hydrodemetallized to obtain a hydrodemetallized product. Preferably, in step (c) the entire mixed deasphalted product as obtained in step (b) is hydrodemetallized.

[0030] The hydrodemetallization of the mixed deasphalted product in step (c) can be achieved by any well known hydrodemetallization treatment wherein the mixed deasphalted product to be hydrodemetallized is passed at elevated temperature and pressure and in the presence of hydrogen in an upward, downward, or radial direction, through one or more vertically disposed reactors containing a fixed or moving bed of hydrodemetallization catalyst particles. In one embodiment, the hydrogen for the hydrodemetallization reactions may be produced, for example, without limitation, by water electrolysis. The water electrolysis process may be powered by renewable energy (such as solar photovoltaic, wind or hydroelectric power) to generate green hydrogen, nuclear energy or by non-renewable power from other sources (grey hydrogen). The hydrodemetallization can be carried out in a bunker flow reactor, a fixed bed reactor, a fixed bed swing reactor or a movable bed reactor. Preferably, the hydrodemetallizing in step (c) is at least partially carried out in a bunker flow or moving bed reactor.

[0031] In step (c), use is made of a hydrodemetallization catalyst. Suitable hydrodemetallization catalysts to be used in accordance with the present invention consist of oxidic carriers such as alumina, silica or silica-alumina, on which one or more Group VIB or Group VIII metals or metal compounds may be deposited. Such hydrodemetallization catalysts are commercially available from many catalyst suppliers. Particularly suitable hydrodemetallization catalysts are those having as the active agent one of the combinations nickel/molybdenum (NiMo) or cobalt/molybdenum (CoMo), optionally promoted with phosphorus (P), on an alumina (Al₂O₃) carrier. Concrete examples of particularly suitable catalysts are CoMo/Al₂O₃, CoMoP/Al₂O₃ and NiMo/Al₂O₃ and NiMoP/Al₂O₃ catalysts.

[0032] The hydrodemetallization in step (c) can suitably be carried out at a hydrogen partial pressure of 20-300 bara, preferably 50-210 bara, a temperature of 300-460° C., preferably 310-435° C., and a space velocity of 0.1-10 hr⁻¹,

preferably 0.2 to 7 hr⁻¹. In some embodiments, the use of the pure and heavy deasphalted product in step (c) will result in considerable metals laydown on the hydrodemetallization catalyst(s) to be used in step (c) which in turn results in a very swift deterioration/deactivation of the hydrodemetallization catalyst(s), requiring a much more regular and fast replacement of the hydrodemetallization catalyst(s) when compared with known processes in which a diluted and lighter deasphalted product with a lower metals content is subjected to a hydrodemetallization step. Hence, the hydrodemetallization reactor is preferably a bunker flow reactor, a fixed bed swing reactor or a movable bed reactor. The hydrodemetallization catalyst(s) is (are) regularly replaced for instance every three weeks or two months, whereas in conventional processes the hydrodemetallization catalyst(s) (like the hydrotreating and hydrocracking catalysts to be used in steps (d) and (e) in the present process) may not be replaced within a year time.

[0033] In step (d), at least part of the hydrodemetallized product as obtained in step (c) is hydrotreated to obtain a hydrotreated product. Preferably, in step (d) the entire hydrodemetallized product as obtained in step (c) is hydrotreated.

[0034] The hydrotreating of the hydrodemetallized product in step (d) can be achieved by any well known hydrotreating process wherein the hydrodemetallized product to be hydrotreated is passed at elevated temperature and pressure and in the presence of hydrogen in an upward, downward or radial direction, through one or more vertically disposed reactors containing a fixed or moving bed of hydrotreating catalyst particles. In one embodiment, the hydrogen for the hydrotreating reactions may be produced, for example, without limitation, by water electrolysis. The water electrolysis process may be powered by renewable energy (such as solar photovoltaic, wind or hydroelectric power) to generate green hydrogen, nuclear energy or by non-renewable power from other sources (grey hydrogen). The hydrotreatment can be carried out in a bunker flow reactor, a fixed bed reactor, a fixed bed swing reactor or a movable bed reactor. Preferably, the hydrotreatment in step (d) is carried out in two reaction zones, whereby the hydrodemetallized product is first passed to a first reaction zone in which the hydrodemetallized product is partly hydrotreated after which the partly hydrotreated effluent so obtained is subjected to further hydrotreatment in a second reaction zone. The first reaction zone and second reaction zone can be arranged in a stacked bed configuration, or the two reactions zones can each be arranged in a separate reactor. Preferably, the first reaction zone and the second reaction zone are respectively arranged in a first reactor and a second reactor. The first reactor may be a bunker flow reactor and the second reactor may be a fixed bed reactor.

[0035] The hydrotreating catalyst to be used in step (d) can suitably be a desulphurization catalyst. The desulphurization catalyst may be any hydrodesulphurization catalyst known in the art. Suitable hydrodesulphurization catalysts comprise a Group VIII metal of the Periodic Table and a compound of a Group VIB metal of the Periodic Table as hydrogenation components on a porous catalyst support, usually alumina or amorphous silica-alumina. Well-known examples of suitable combinations of hydrogenation compounds are cobalt-molybdenum, nickel-molybdenum, nickel-tungsten, and nickel-cobalt-molybdenum. A hydrodesulphurization catalyst comprising compounds of nickel and/or cobalt and molybdenum

as hydrogenation compounds is preferred. The hydrodesulphurization catalyst may further comprise a cracking component such as for example Y zeolite. It is, however, preferred that no substantial hydrocracking takes place in the hydrotreatment in step (d). Therefore, it is preferred that the catalyst is substantially free of a cracking component. A catalyst comprising nickel and/or cobalt and molybdenum supported on alumina without a zeolitic cracking compound is particularly preferred.

[0036] The hydrotreating conditions in step (d), i.e., temperature, pressure, hydrogen supply rate, weight hourly velocity of the feedstock, are typical hydrotreating conditions. Preferably, the temperature used for the hydrotreating in step (c) is in the range of from 280 to 430° C., more preferably in the range of from 320 to 420° C., and most preferably in the range of from 330 to 410° C.

[0037] Suitable hydrotreating pressures are in the range of from 10 to 300 bara. Preferably, the hydrotreating pressure is in the range of from 30 to 250 bara, more preferably in the range of from 80 to 220 bara.

[0038] In step (e), at least part of the hydrotreated product as obtained in step (d) is hydrocracked to obtain a hydrocracked product. Preferably, in step (e) the entire hydrotreated product as obtained in step (d) is hydrocracked.

[0039] The hydrocracking in step (e) of the process according to the present invention may be conducted in any way known in the art, provided that at least one of the catalysts used in the hydrocracking zone is acidic. Suitably, the hydrocracking is carried out in the presence of hydrogen and a suitable hydrocracking catalyst at elevated temperature and pressure. In one embodiment, the hydrogen for the hydrocracking reactions may be produced, for example, without limitation, by water electrolysis. The water electrolysis process may be powered by renewable energy (such as solar photovoltaic, wind or hydroelectric power) to generate green hydrogen, nuclear energy or by non-renewable power from other sources (grey hydrogen). Suitable hydrocracking catalysts consist of one or more metals from nickel, tungsten, cobalt and molybdenum in elemental, oxidic or sulphidic form on a suitable carrier such as alumina, silica, silica-alumina or a zeolite. There are many commercially available hydrocracking catalysts which can be suitably applied in the process of the present invention. At least one of the catalysts used in the hydrocracking zone must be acidic, i.e., must contain a silica-alumina and/or zeolitic component.

[0040] The hydrocracking in step (e) can be carried out in a single- or multiple-stage mode of operation. In the case of a single-stage mode of operation, a stacked bed of a hydrodenitrification/first-stage hydrocracking catalyst on top of a conversion catalyst can suitably be used. Particularly suitable hydrodenitrification/first-stage hydrocracking catalysts are NiMo/Al₂O₃ and CoMo/Al₂O₃, optionally promoted with phosphorus and/or fluor. Preferred conversion catalysts are those based on NiW/zeolite or NiW/zeolite/silica-alumina. Suitable hydrocracking conditions in step (e) are an operating pressure of 80-250 bara, preferably 90-220 bara, and a temperature of 300-460° C., preferably 350-430° C.

[0041] Without being bound by theory, it has been surprisingly found that by introducing a hydrogen deficient feed as a portion of the mixed deasphalted product only the activity of the catalyst in hydrodemetallization reactors increases. For example, in a simulation of including of slurry oil (SLO) in the deasphalted oil (DAO) to form a mixed

DAO feed, the average hydrogen content of the DAO feed was reduced by 0.3% wt and the exothermicity over the hydrodemetallization catalyst increased by 20% whilst the exothermicity over the following hydrotreating and hydrocracking catalyst remained nearly constant. This increase in activity of the hydrodemetallization catalyst allowed a reduction of the inlet temperature of the (first) hydrometallization reactor enabling a reduction of the duty of the reactor feed furnace to the hydrodemetallization reactor, which in turn may be used for an increase of the cycle length of the unit.

[0042] In step (f), at least part of the hydrocracked product as obtained in step (e) is subjected to a separation treatment to obtain at least a middle distillate fraction. Preferably, in step (f) the entire hydrocracked product as obtained in step (e) is subjected to the separation treatment.

[0043] The separation treatment is step (f) can suitably a fractionating treatment which is carried out at a temperature in the range from 50 to 400° C., preferably at a temperature in the range of from 70 to 370° C., and a pressure in the range of from 0.03 to 15 bara, preferably a pressure in the range of from 0.05 to 10 bara.

[0044] From step (f), a fuel gas fraction may also be obtained, preferably containing C1-C2 products. In other embodiments, the fuel gas fraction may include C1-C4 products. In some embodiments, if a renewable hydrogen deficient feed is used in step (b), and at least part of the fuel gas fraction will have a renewable origin. Thus, if the fuel gas having a renewable origin is used in hydrogen production, a portion of the hydrogen produced may be considered as having a renewable origin.

[0045] Beside of the middle distillate fraction to be obtained in step (f) there can also be obtained a heavy residual fraction. Suitably, at least 80 wt % of the heavy residual fraction also obtained in the separation treatment in step (f) has a boiling point above 370° C. Preferably, at least 90% of the heavy residual fraction also obtained in the separation treatment in step (f) has a boiling point above 370° C.

[0046] At least part of a heavy residual fraction also obtained in step (f) may be recycled to step (a). In this way an improved yield of middle distillates can be obtained.

[0047] Alternatively, said heavy fraction could also be suitably applied as a feed for a fluidised bed catalytic cracking (FCC) unit or as a feedstock for lubricating oil manufacture. Of course, a combination of these options is possible as well.

[0048] In order to achieve an optimum middle distillates yield, it is preferred that at least a part of the heavy fraction obtained in step (f) is again subjected to hydrocracking to improve the yield of middle distillates. Hence, in a preferred embodiment at least part of a heavy residual fraction which is also obtained in step (f) is recycled to step (e).

[0049] In another preferred embodiment of the present invention at least part of the heavy residual fraction also obtained in step (f) is recycled to step (a) and at least part of the heavy residual fraction also obtained in step (f) is recycled to step (e). In this way the yield of middle distillates is further improved.

[0050] In yet another preferred embodiment, at least part of a heavy residual fraction also obtained in step (f) is subjected to a further hydrocracking step (g), and at least part of the hydrocracked product as obtained in such a step

(g) is recycled to step (f). Also, this embodiment ensures that an optimal yield of middle distillates will be established.

[0051] Preferably, at least part of the heavy residual fraction also obtained in step (f) is also recycled to step (a) to improve the middle distillate yield even further.

[0052] Preferably, the hydrocracking in step (e) and/or step (g) is carried out in two or more reaction zones. Preferably, the two or more reaction zones are arranged in a stacked bed configuration.

[0053] Preferably, at least part of the asphaltic product as obtained in step (a) is subjected to a gasification step (h) to obtain hydrogen and carbon monoxide. In some embodiments, if a renewable hydrogen deficient feed is used in step (b), and at least part of the heavy residual fraction also obtained in step (f) is recycled to step (a), the asphaltic product as obtained in step (a) would have a portion of renewable origin.

[0054] The asphaltic product as obtained in step (a) may be used in several ways. It can for instance be combusted for cogeneration of power and steam. Alternatively, it can be partially combusted for clean fuel gas production, cogeneration of power and steam, hydrogen manufacture or hydrocarbon synthesis. In another embodiment, if the asphalt has a portion of renewable origin, the hydrogen manufactured therefrom would be considered to be partially of green origin. Still another option is application in bitumen, emulsion fuels or solid fuels by means of pelletizing.

[0055] Preferably, such a gasification step (h) is a partial combustion step.

[0056] In a preferred embodiment at least part of the hydrogen as obtained in step (h) is recycled to at least one of steps (c), (d), (e) and (g).

[0057] The middle distillate fraction as obtained in step (f) comprises middle distillates which contain less than 10 ppmwt of sulphur. Preferably, the middle distillates contain less than 8 ppmwt of sulphur, more preferably less than 6 ppmwt of sulphur, and most preferably less than 5 ppmwt of sulphur.

[0058] FIG. 1 depicts the process according to the present invention; FIG. 2 depicts another embodiment of the present invention; and FIG. 3 depicts a further embodiment of the process according to the present invention.

[0059] In FIG. 1 an atmospheric or reduced pressure hydrocarbon oil residue is passed via a line 1 into a deasphalting unit 2 in which a deasphalted product and an asphaltic product are obtained. A hydrogen deficient feed is fed via line 20 to combine with at least part of the deasphalted product in line 3 to form a mixed deasphalted product. At least part of the mixed deasphalted product is passed into a hydrodemetallization unit 5 and the asphaltic product is withdrawn from the deasphalting unit 2 via a line 4. At least part of the hydrodemetallized product as obtained in hydrodemetallization unit 5 is passed via a line 6 to hydrotreating unit 7. At least part of the hydrotreated product as obtained in the hydrotreating unit 7 is then passed to a hydrocracking unit 9 via a line 8. At least part of the hydrocracked product as obtained in the hydrocracking unit 9 is passed via a line 10 to a fractionating unit 11 from which at least a middle distillate fraction is recovered via a line 12.

[0060] FIG. 2 is an extension of FIG. 1 in that in the fractionating unit 11 also a heavy residual fraction is obtained which is withdrawn via a line 13 from the fractionating unit 11 and at least part of the heavy residual fraction is recycled to the deasphalting unit 2.

[0061] FIG. 3 is an extension of FIG. 2 in that in the fractionating unit 11 also a heavy residual fraction is obtained which is withdrawn via a line 13 from the fractionating unit 11 and at least part of the heavy residual fraction is recycled via a line 14 to the hydrocracking unit 9 and/or at least part of the heavy residual fraction is recycled via a line 15 to the hydrodemetallization unit 5 and/or at least part of the heavy residual fraction is recycled via a line 16 to the deasphalting unit 2.

[0062] FIG. 4 is an extension of FIG. 1 in that in the fractionating unit 11 also a heavy residual fraction is obtained which is withdrawn via a line 13 from the fractionating unit 11 and at least part of the heavy residual fraction is passed via the line 13 to a hydrocracking unit 17 from which at least a distillate fraction is recovered via a line 21. At least part of the hydrocracked product as obtained in the hydrocracking unit 14 is recycled to fractionating unit 11 via lines 18 and 19 and at least part of the heavy residual fraction as obtained in the fractionating unit 11 is recycled via lines 18 and 20 to the deasphalting unit 2.

[0063] Although a few embodiments of the disclosure have been described in detail above, those of ordinary skill in the art will readily appreciate that many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

1. A process for producing middle distillates from a feedstock comprising a residual hydrocarbonaceous feedstock and a hydrogen deficient feedstock, comprising the steps of:

- (a) deasphalting the residual hydrocarbonaceous feedstock to obtain a deasphalted product of which at least 50 wt % has a boiling point above 550° C. and an asphaltic product;
- (b) combining the deasphalted product with the hydrogen deficient feedstock to produce a mixed deasphalted product, wherein the hydrogen deficient feedstock has a hydrogen (H) content of at least 6 wt % to at most 11.3 wt %;

(c) hydrodemetallizing at least part of the mixed deasphalted product from step (b) to produce a hydrodemetallized product;

(d) hydrotreating at least part of the hydrodemetallized product from step (c) to produce a hydrotreated product;

(e) hydrocracking at least part of the hydrotreated product from step (d) to produce a hydrocracked product; and

(f) subjecting at least part of the hydrocracked product from step (e) to a separation treatment to produce at least a middle distillate fraction.

2. The process according to claim 1, wherein the amount of hydrogen deficient feedstock comprises up to about 50 wt % of the mixed deasphalted product.

3. The process according to claim 1, wherein the mixed deasphalted product has a hydrogen content less than at least 0.1 wt % of the deasphalted product via the addition of the hydrogen deficient feedstock.

4. The process according to claim 1, wherein at least a portion of the hydrogen deficient feedstock includes a portion of renewable origin.

5. The process according to claim 1, wherein the hydrogen deficient feedstock comprises slurry oil from a fluid catalytic cracker (FCC) unit.

6. A process according to claim 4, wherein the hydrogen deficient feedstock comprises tall oil pitch from the distillation of crude tall oil.

7. A process according to claim 1, wherein the middle distillate fraction in produced step (f) contain less than 10 ppmwt of sulphur.

8. A process according to claim 1, wherein the separation treatment in step (f) produces a heavy residual fraction of which 80 wt % has a boiling point above 370° C.

9. A process according to claim 1, wherein in the separation treatment in step (f) also produces a fuel gas fraction having a portion of renewable origin.

10. A process according to claim 1, wherein at least part of the asphaltic product as obtained in step (a) is subjected to a gasification step (h) to obtain hydrogen having a portion of renewable origin and carbon monoxide.

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